

Hemanth Katikala Muniraj

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Summary

Robotics Engineer with hands-on experience developing autonomous systems, industrial robotics, and AI-driven automation across autonomous vehicles, robotic manipulation, and computer vision. Experienced in designing simulation workflows, robotics software, perception pipelines, motion planning, and industrial automation using ROS, MATLAB/Simulink, Python, and C++. Built end-to-end robotic solutions involving mobile robots, industrial manipulators, machine learning, sensor fusion, and autonomous navigation while improving reliability, deployment efficiency, and system validation. Strong background collaborating across software, infrastructure, controls, and AI teams to deliver scalable robotics solutions from research through production-oriented deployments.

Skills

- **Programming Languages:** Python, C++, MATLAB, SQL, Bash, Java, Embedded C
- **Robotics Frameworks:** ROS, ROS2, MoveIt, Nav2, Gazebo, RViz, CoppeliaSim, URDF, TF, Xacro
- **Autonomous Systems:** SLAM, Localization, Path Planning, Motion Planning, Autonomous Navigation, Behavior Planning, Trajectory Generation, Occupancy Grid Mapping
- **Perception & AI:** Computer Vision, OpenCV, YOLO, PyTorch, TensorFlow, CNNs, Object Detection, Image Segmentation, 3D Object Recognition, OCR
- **Industrial Robotics:** Universal Robots (UR3/UR5/UR10), ABB RobotStudio, FANUC (Fundamentals), Robotic Manipulation, Robot Calibration, TCP Calibration, Coordinate Frames
- **Simulation:** MATLAB, Simulink, Gazebo, CoppeliaSim, Autonomous Vehicle Simulation, Log Replay Simulation, Scenario-Based Testing
- **Controls:** PID Control, State Estimation, Kalman Filter, Extended Kalman Filter, Kinematics, Inverse Kinematics, Dynamics
- **Sensors:** RGB Camera, Stereo Camera, LiDAR, IMU, GPS, Wheel Encoder, Depth Camera
- **Automation & Infrastructure:** Airflow, Git, GitHub, Linux, Ubuntu, Docker, MQTT, REST APIs, CI/CD, YAML
- **Cloud & AI Platforms:** Azure Machine Learning, MLflow, Azure ML SDK v2, Model Deployment, Compute Clusters
- **Engineering Tools:** SolidWorks, Fusion 360, Arduino, ESP32, Raspberry Pi, PLC Ladder Logic, SCADA

Work Experience

Infrastructure Intern

Las Vegas, NV, USA

Motional

May 2026 – Present

- Developed and validated autonomous vehicle simulation scenarios using Motional's LogSim framework by creating YAML-based scenario configurations to evaluate planner behavior across pedestrian, cyclist, and vehicle interaction scenarios.
- Automated scenario generation and parameter sweep workflows using Python, Airflow, and internal infrastructure tools, significantly reducing manual scenario configuration while improving repeatability of large-scale validation experiments.
- Investigated and resolved simulation discrepancies by debugging replay synchronization, trigger conditions, agent behaviors, warm-start configurations, map alignment, and planner interactions to improve simulation fidelity.
- Built configurable scenario variants through log replay, synthetic agent generation, and scenario fuzzing to expand regression test coverage for autonomous driving software across diverse traffic conditions.
- Collaborated with autonomy, infrastructure, and simulation engineers to reproduce planner edge cases, analyze telemetry, document experimental findings, and deliver production-quality validation workflows.

Industrial Automation and Robotics Intern

AliveMind

Cerritos, CA, USA

Sep 2025 – Dec 2025

- Led the robotics and AI development for an autonomous Robot Barista system by integrating a Universal Robots UR5 manipulator with perception, voice interaction, and task-level automation for beverage preparation.
- Developed modular ROS-based software integrating robot motion planning, MQTT communication, CoppeliaSim simulation, Baxter gripper control, face recognition, and speech recognition to enable end-to-end robotic task execution.
- Implemented computer vision and AI pipelines using OpenCV, Vosk speech recognition, and facial recognition to support customer identification, voice-controlled ordering, and autonomous robot decision making.
- Designed, tested, and optimized robotic manipulation workflows through simulation and hardware validation, improving task execution reliability, system modularity, and integration across multiple software components.
- Collaborated with cross-functional robotics, controls, and AI teams to integrate Simulink-based Driver Monitoring System communication, debug hardware-software interactions, and accelerate deployment of intelligent automation solutions.

Computer Vision and 3D Perception Intern

BotClub Pvt Ltd

Visakhapatnam, India

Jun 2023 – Nov 2023

- Developed deep learning-based object detection and 3D perception pipelines using PyTorch, OpenCV, and Ultralytics YOLO to improve automated visual inspection accuracy.
- Built computer vision models capable of recognizing and classifying industrial components from 2D images and 3D STL models to support robotic perception applications.
- Optimized data preprocessing, augmentation, and model training workflows, improving object detection performance while reducing false positives across diverse datasets.
- Integrated AI-based perception outputs into augmented reality visualization workflows, enabling interactive inspection and visualization of detected objects.
- Collaborated with AI engineers to evaluate model performance, troubleshoot perception pipelines, and deliver scalable computer vision solutions for industrial automation.

Research Fellow – Embedded Systems and Robotics

Vellore Institute of Technology

Amaravathi, India

Dec 2023 – Apr 2024

- Conducted research on Fog-assisted IoT architectures for smart agriculture, resulting in the publication of the AGRIFOG system in IEEE Xplore.
- Designed distributed IoT communication pipelines using MQTT, ESP32, Raspberry Pi, and cloud services to enable reliable real-time environmental monitoring.
- Developed REST API and Google Drive integration for automated sensor data collection, storage, and visualization across edge computing devices.
- Built embedded monitoring systems capable of collecting temperature, humidity, soil moisture, and environmental data to support precision agriculture applications.
- Collaborated with faculty researchers to validate system performance, analyze experimental results, and prepare technical documentation for peer-reviewed publication.

Projects

Autonomous Drone Navigation and Dynamic Obstacle Avoidance *MATLAB, UAV Toolbox, Computer Vision, Control Systems*

- Developed a complete autonomous drone simulation capable of waypoint navigation, dynamic obstacle avoidance, wind disturbance compensation, and first-person (FPV) visual feedback in MATLAB.
- Designed a vision-based collision avoidance pipeline using onboard camera perception to detect moving obstacles and generate safe local trajectories while maintaining mission objectives.
- Implemented waypoint guidance, PID-based flight control, and adaptive trajectory replanning to ensure stable navigation under changing environmental conditions.
- Built telemetry logging, mission replay, and video recording modules to analyze vehicle behavior, evaluate controller performance, and support autonomous system validation.

Robot Barista Automation System *ROS, CoppeliaSim, UR5, MQTT, OpenCV, Python*

- Developed an autonomous robotic barista using a Universal Robots UR5 manipulator capable of beverage preparation through integrated motion planning, perception, and task automation.
- Integrated MQTT communication, ROS nodes, Baxter gripper control, and CoppeliaSim simulation to coordinate robotic manipulation and system-level task execution.
- Implemented face recognition and speech recognition pipelines using OpenCV and Vosk to enable customer identification and voice-controlled beverage ordering.
- Designed a modular robotics software architecture supporting perception, planning, robot control, and human-machine interaction, enabling scalable development and simplified debugging.

Autonomous Warehouse Navigation Robot *ROS2, Nav2, Gazebo, RViz, LiDAR*

- Built a fully autonomous mobile robot capable of localization, mapping, path planning, and obstacle avoidance using the ROS2 Navigation Stack.
- Integrated SLAM, AMCL localization, costmaps, recovery behaviors, and LiDAR perception to achieve reliable navigation in dynamic warehouse environments.
- Tuned global and local planners to improve navigation efficiency while achieving localization accuracy within approximately ± 5 cm.
- Validated robot performance in Gazebo through repeated navigation experiments involving dynamic obstacles, waypoint following, and autonomous recovery behaviors.

FactoryVision-AI Intelligent Inspection System *Python, OpenCV, YOLO, Streamlit*

- Developed an AI-powered industrial quality inspection platform capable of detecting manufacturing defects from uploaded images and live production simulations.
- Designed an interactive factory dashboard featuring production monitoring, defect visualization, inspection analytics, and real-time AI inference.
- Integrated YOLO-based object detection with computer vision pipelines to automate visual inspection and support intelligent manufacturing workflows.
- Engineered a modular software architecture enabling scalable deployment, rapid inference, and future integration with robotic sorting and industrial automation systems.

Traffic Sign Detection using YOLOv11 *Python, PyTorch, YOLOv11, OpenCV*

- Developed a deep learning-based traffic sign detection system using YOLOv11 to classify 10 traffic sign categories under diverse driving conditions.
- Performed dataset preprocessing, augmentation, and model optimization to improve robustness against rain, snow, low-light, and motion blur.
- Trained and evaluated the model on over 1,200 labeled images using PyTorch, achieving reliable real-time inference suitable for autonomous driving applications.
- Optimized the inference pipeline for deployment by improving detection speed, reducing false positives, and increasing robustness across challenging environmental scenarios.

Education

University of Michigan, Dearborn

Sep 2024 – May 2026

M.S.E. Robotics Engineering

GPA: 3.96/4.0

Vellore Institute of Technology, Amaravathi

Sep 2020 – May 2024

B.Tech Computer Science (Robotics Specialization)

GPA: 3.48/4.0

Certifications: Universal Robots e-Series, ABB RobotStudio, PLC Ladder Logic, Geometric Dimensioning and Tolerancing (GD&T)